

Vectors, Vector Spaces and Linear Algebra

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Sets and Algebraic Structures

Set: A well-defined collection of objects.

$$A = \{1, 2, 3, 4\}$$

Important operations:

$$A \cup B, \quad A \cap B, \quad A - B, \quad A^c, \quad A \times B$$

An algebraic structure is a set with a binary operation:

$$(S, *)$$

Structure	Additional property
Magma	Closure
Semigroup	Closure + Associativity
Monoid	Semigroup + Identity
Group	Monoid + Inverse
Abelian Group	Group + Commutativity

ML connection: Sets describe datasets/classes, while algebraic structures provide the mathematical foundation for transformations and operations.

Vectors and Vector Spaces

A vector is an ordered collection of numbers:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}, \quad \|\mathbf{v}\| = \sqrt{\sum_i v_i^2}.$$

Vector addition:

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \end{bmatrix}$$

Scalar multiplication:

$$\alpha \mathbf{v} = \begin{bmatrix} \alpha v_1 \\ \alpha v_2 \\ \vdots \end{bmatrix}$$

A vector space is closed under vector addition and scalar multiplication.

Linear Combinations, Hyperplanes and Perceptron

A linear combination is

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k.$$

The equation

$$ax + by + c = 0$$

represents a line, while

$$\mathbf{w}^T \mathbf{x} + b = 0$$

represents a hyperplane.

$\mathbf{w} = \text{normal vector to the hyperplane}$

A perceptron computes

$$z = \mathbf{w}^T \mathbf{x} + b.$$

Using

$$\mathbf{w}^T \mathbf{x} = \|\mathbf{w}\| \|\mathbf{x}\| \cos \theta,$$

the angle between $\mathbf{w}$ and $\mathbf{x}$ influences classification.

Mathematical Spaces

Different mathematical spaces provide different ways to study mathematical objects.

Space	Main concept
Vector Space	Addition and scalar multiplication
Metric Space	Distance
Inner Product Space	Angle and orthogonality
Hilbert Space	Complete inner product space
Topological Space	Continuity and convergence

For an inner product:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v}$$

Angle:

$$\cos \theta = \frac{\mathbf{u}^T \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

AI/ML applications:

Metric $\rightarrow$ KNN/Clustering

Inner Product $\rightarrow$ Similarity

Vectors

A vector is a mathematical object having magnitude and direction.

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

Examples:

$$\mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}$$

Magnitude:

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

In machine learning, a data point can be represented as a feature vector:

$$\mathbf{x} = [x_1, x_2, \dots, x_n]^T.$$

Coordinate System

A coordinate system provides a way to represent points and vectors.

2D:

$$P = (x, y), \quad \mathbf{p} = \begin{bmatrix} x \\ y \end{bmatrix}$$

3D:

$$P = (x, y, z), \quad \mathbf{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Standard basis vectors in $\mathbb{R}^3$:

$$\mathbf{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Thus:

$$\mathbf{v} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}.$$

Vector Addition

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix},$$

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$$

Example:

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

Geometrically, vector addition follows the parallelogram law.

Resultant = diagonal of parallelogram

Scalar Multiplication

For scalar α and vector $\mathbf{v}$:

$$\alpha \mathbf{v} = \begin{bmatrix} \alpha v_1 \\ \alpha v_2 \\ \vdots \\ \alpha v_n \end{bmatrix}$$

Example:

$$3 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

Magnitude:

$$\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\|$$

- $\alpha > 1$: vector becomes longer
- $0 < \alpha < 1$: vector becomes shorter
- $\alpha < 0$: direction reverses
- $\alpha = 0$: zero vector

Linear Combinations

A linear combination of vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ is

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n.$$

Example:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$3\mathbf{v}_1 + 2\mathbf{v}_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Therefore every vector in $\mathbb{R}^2$ can be expressed as a linear combination of the standard basis vectors.

Span

The span of vectors is the set of all their linear combinations.

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \left\{ \sum_{i=1}^n c_i \mathbf{v}_i : c_i \in \mathbb{R} \right\}$$

For

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

$$\text{span}\{\mathbf{e}_1, \mathbf{e}_2\} = \mathbb{R}^2.$$

Geometrically:

one independent vector $\rightarrow$ line

two independent vectors in $\mathbb{R}^2 \rightarrow$ plane.

Basis Vectors

A basis is a set of vectors that:

- ① spans the vector space
- ② is linearly independent

Standard basis of $\mathbb{R}^2$:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Every vector has a unique representation:

$$\mathbf{v} = x\mathbf{e}_1 + y\mathbf{e}_2.$$

For example:

$$\begin{bmatrix} 4 \\ 3 \end{bmatrix} = 4\mathbf{e}_1 + 3\mathbf{e}_2.$$

Matrices as Linear Transformations

A matrix can represent a linear transformation.

$$A : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$\mathbf{x} \mapsto A\mathbf{x}$$

Example:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$A\mathbf{x} = \begin{bmatrix} 2x \\ 3y \end{bmatrix}$$

The transformation scales the x -direction by 2 and the y -direction by 3.

Matrix Multiplication as Composition

Suppose

$$\mathbf{x} \xrightarrow{B} B\mathbf{x} \xrightarrow{A} A(B\mathbf{x}).$$

Therefore:

$$A(B\mathbf{x}) = (AB)\mathbf{x}.$$

Hence:

$$AB = \text{composition of } B \text{ followed by } A$$

Matrix multiplication is generally not commutative:

$$AB \neq BA.$$

This reflects the fact that applying transformations in different orders generally gives different results.

Three-Dimensional Linear Transformations

A linear transformation in 3D maps:

$$\mathbb{R}^3 \rightarrow \mathbb{R}^3.$$

Common transformations:

- Scaling
- Rotation
- Reflection
- Shearing

Scaling:

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2x \\ 3y \\ z \end{bmatrix}.$$

3D Rotation

Rotation around the z-axis:

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The transformed vector is:

$$\mathbf{x}' = R_z(\theta)\mathbf{x}.$$

Applications:

- Computer graphics
- Computer vision
- Robotics
- 3D image processing

Determinant

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$\det(A) = ad - bc$$

Example:

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

$$\det(A) = 2(4) - 3(1) = 5.$$

Geometric interpretation:

$$|\det(A)| = \text{area scaling factor}$$

For a 3D transformation:

$$|\det(A)| = \text{volume scaling factor.}$$

If

Inverse Matrices

The inverse reverses a matrix transformation:

$$A^{-1}A = AA^{-1} = I.$$

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

provided:

$$ad - bc \neq 0.$$

Therefore:

$$\det(A) \neq 0 \iff A^{-1} \text{ exists}$$

for a square matrix.

Column Space

The column space is the span of the columns of a matrix.

$$\text{Col}(A) = \text{span}\{\text{columns of } A\}$$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Since

$$\begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

$$\text{Col}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}.$$

The column space represents all possible outputs $A\mathbf{x}$.

Null Space

The null space contains all vectors mapped to the zero vector.

$$\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$$

For

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

solve:

$$x + 2y = 0.$$

Thus:

$$x = -2y$$

and

$$\mathbf{x} = t \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Therefore:

$$\text{Null}(A) = \left\{ t \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}$$

Nonsquare Matrices as Transformations

A matrix need not be square.

For example:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$$

is a 2×3 matrix.

Therefore:

$$A : \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

and

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x + 2z \\ y + 3z \end{bmatrix}.$$

In general:

$$m \times n \text{ matrix} : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

This is useful for dimensionality reduction and feature transformations.

Dot Product

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix},$$

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i$$

or:

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}.$$

Geometric form:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

If:

$$\mathbf{u} \cdot \mathbf{v} = 0,$$

the vectors are orthogonal.

Dot Product and Duality

A vector can define a linear functional.

For

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix},$$

define:

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}.$$

Thus:

$$f(x, y) = w_1 x + w_2 y.$$

The set of linear functionals on a vector space forms its dual space.

$$\boxed{\mathbf{w} \longrightarrow \mathbf{w}^T(\cdot)}$$

This concept is important in linear algebra, optimization and machine learning.

Cross Product

For vectors in $\mathbb{R}^3$:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix},$$

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}.$$

The result is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$.

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0$$

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0$$

Cross Product: Geometric Meaning

Magnitude:

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta$$

This equals the area of the parallelogram formed by $\mathbf{u}$ and $\mathbf{v}$.

$$A_{\text{parallelogram}} = \|\mathbf{u} \times \mathbf{v}\|$$

Triangle area:

$$A_{\text{triangle}} = \frac{1}{2} \|\mathbf{u} \times \mathbf{v}\|$$

The cross product therefore provides a normal vector and measures area.

Cross Product and Linear Transformations

Let:

$$\mathbf{n} = \mathbf{u} \times \mathbf{v}.$$

Then $\mathbf{n}$ is normal to the plane containing $\mathbf{u}$ and $\mathbf{v}$.

A plane can be written as:

$$\mathbf{n}^T(\mathbf{x} - \mathbf{x}_0) = 0.$$

Equivalently:

$$\boxed{\mathbf{n}^T \mathbf{x} = c}$$

For an invertible 3×3 matrix:

$$\boxed{(\mathbf{A}\mathbf{u}) \times (\mathbf{A}\mathbf{v}) = \det(\mathbf{A})\mathbf{A}^{-T}(\mathbf{u} \times \mathbf{v})}$$

This connects cross products with determinants, inverse transformations, area and orientation.

Cramer's Rule

For:

$$A\mathbf{x} = \mathbf{b},$$

Cramer's rule uses determinants to solve the system.

For:

$$a_1x + b_1y = c_1, \quad a_2x + b_2y = c_2,$$

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

$$x = \frac{D_x}{D}, \quad y = \frac{D_y}{D}.$$

The determinant D_x replaces the first column of A by $\mathbf{b}$, while D_y replaces the second column.

Cramer's Rule: Example

Solve:

$$2x + y = 5$$

$$x - y = 1.$$

$$D = \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3$$

$$D_x = \begin{vmatrix} 5 & 1 \\ 1 & -1 \end{vmatrix} = -6$$

$$x = \frac{-6}{-3} = 2.$$

$$D_y = \begin{vmatrix} 2 & 5 \\ 1 & 1 \end{vmatrix} = -3$$

$$y = \frac{-3}{-3} = 1.$$

Change of Basis

The same vector can have different coordinate representations in different bases.

Standard basis:

$$E = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

Another basis:

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

For:

$$\mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix},$$

we find:

$$\mathbf{v} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Thus:

$$[\mathbf{v}]_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Change-of-Basis Matrix

Put the basis vectors into the columns of P :

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Then:

$$[\mathbf{v}]_E = P[\mathbf{v}]_B.$$

Therefore:

$$\boxed{[\mathbf{v}]_B = P^{-1}[\mathbf{v}]_E}$$

The vector remains the same geometrically; only its coordinate representation changes.

Eigenvectors and Eigenvalues

An eigenvector is a vector whose direction remains unchanged under a linear transformation.

$$A\mathbf{v} = \lambda\mathbf{v}$$

where:

- $\mathbf{v}$ = eigenvector
- λ = eigenvalue

The transformation only scales the eigenvector.

If $\lambda > 1$, it expands.

If $0 < \lambda < 1$, it contracts.

If $\lambda < 0$, its direction reverses.

Finding Eigenvalues

Starting with:

$$A\mathbf{v} = \lambda\mathbf{v}$$

$$(A - \lambda I)\mathbf{v} = 0.$$

For a nonzero $\mathbf{v}$:

$$\det(A - \lambda I) = 0.$$

This is the characteristic equation.

Example:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

$$\det(A - \lambda I) = (2 - \lambda)(3 - \lambda) = 0.$$

Hence:

$$\boxed{\lambda_1 = 2, \quad \lambda_2 = 3}$$

Eigenvector Example

For:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix},$$

for $\lambda = 2$:

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Thus:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

For $\lambda = 3$:

$$A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Thus:

$$\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Applications of Eigenvalues and Eigenvectors

Eigenvalues and eigenvectors are used in:

- Principal Component Analysis (PCA)
- Image processing
- Dimensionality reduction
- Dynamical systems
- Differential equations
- Google PageRank
- Quantum mechanics

In PCA, eigenvectors of the covariance matrix identify important directions of variation in the data.

Vector Spaces

A vector space is a set closed under vector addition and scalar multiplication and satisfying the vector-space axioms.

Examples:

$$\mathbb{R}^2, \quad \mathbb{R}^3, \quad \mathbb{R}^n$$

Polynomial vector space:

$$P_2 = \{a + bx + cx^2 : a, b, c \in \mathbb{R}\}.$$

A polynomial:

$$p(x) = a + bx + cx^2$$

can be represented as:

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Therefore:

$$P_2 \cong \mathbb{R}^3$$

Vectors and the Perceptron

A perceptron receives an input vector:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

and has a weight vector:

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}.$$

The weighted sum is:

$$z = \mathbf{w}^T \mathbf{x} + b.$$

Perceptron and Angle Between Vectors

Using the dot product:

$$\mathbf{w}^T \mathbf{x} = \|\mathbf{w}\| \|\mathbf{x}\| \cos \theta.$$

Therefore:

$$z = \|\mathbf{w}\| \|\mathbf{x}\| \cos \theta + b.$$

Ignoring b :

$$\theta < 90^\circ \Rightarrow \mathbf{w}^T \mathbf{x} > 0$$

$$\theta = 90^\circ \Rightarrow \mathbf{w}^T \mathbf{x} = 0$$

$$\theta > 90^\circ \Rightarrow \mathbf{w}^T \mathbf{x} < 0.$$

Thus the angle between the input vector and weight vector influences classification.

Complete Conceptual Connection

Vectors $\rightarrow$ Linear Combinations $\rightarrow$ Span $\rightarrow$ Basis $\rightarrow$ Vector Spaces

Matrices $\rightarrow$ Linear Transformations $\rightarrow$ Composition $\rightarrow \det(A) \rightarrow A^{-1}$

Dot Product $\rightarrow$ Angle $\rightarrow$ Orthogonality $\rightarrow$ Duality

Eigenvectors $\rightarrow$ Special Directions $\rightarrow$ PCA and ML

$\mathbf{w}^T \mathbf{x} + b = 0 \rightarrow$ Hyperplane $\rightarrow$ Perceptron

Summary

- Vectors represent quantities with magnitude and direction.
- Linear combinations generate spans and lead to basis vectors.
- Matrices represent linear transformations.
- Matrix multiplication represents composition of transformations.
- Determinants measure area or volume scaling and determine invertibility.
- Column space describes possible outputs; null space describes vectors mapped to zero.
- Dot products measure alignment and provide angles and orthogonality.
- Cross products produce normal vectors and measure area in $\mathbb{R}^3$.
- Cramer's rule solves linear systems using determinants.
- Change of basis changes coordinate representation without changing the vector itself.
- Eigenvectors identify directions preserved by a transformation.
- Vector spaces provide the mathematical framework for all these concepts.

Thank You